# Immune Cell Profiling of Peripheral Blood Using Flow Cytometry and Dimensionality Reduction

A Computational Analysis of Myeloid and Lymphoid Populations in the Context of COVID-19 Immune Surveillance

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# Abstract

*Rapid and accurate identification of immune cell populations in peripheral blood is critical for understanding patient immune responses, particularly in infectious diseases such as COVID-19. In this project, I applied principal component analysis (PCA) and t-distributed stochastic neighbour embedding (t-SNE) to a flow cytometry dataset comprising 1,000 cells measured across 21 surface markers. The dataset was derived from a healthy control sample published in a longitudinal COVID-19 immune profiling study. My analysis successfully identified eight distinct immune cell populations, including neutrophils, T cells, monocytes, NK cells, eosinophils, basophils, B cells, and dendritic cells. PCA revealed that eight principal components were sufficient to capture over 80% of the total variance, with PC1 primarily driven by granularity markers (SSC-A, SSC-H) and CD10, separating neutrophils from lymphocytes, and PC2 driven by HLA-DR and CD14, isolating monocytes. t-SNE with a perplexity of 50 produced the clearest cluster separation across all eight populations. These findings demonstrate that computational dimensionality reduction can serve as a scalable, reproducible alternative to manual gating in clinical flow cytometry workflows, with direct implications for immune monitoring in COVID-19 patients and broader diagnostic applications.*


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# 1. Introduction

The COVID-19 pandemic exposed significant gaps in our ability to rapidly characterise patient immune responses at scale. While severe cases were associated with dysregulated myeloid responses and lymphopenia, clinicians needed tools to quickly profile immune cell populations in patient blood samples to guide treatment decisions. Flow cytometry has long been the gold standard for immune cell phenotyping, but traditional manual gating analysis is time-consuming, subjective, and difficult to standardise across clinical sites.

In this project, I applied unsupervised machine learning techniques to a flow cytometry dataset from a published COVID-19 immune profiling study (Lucas et al., 2020). The original study tracked immune signatures across 113 COVID-19 patients and healthy controls, revealing that key myeloid cell populations were predictive of disease severity. I focused on a subset of 1,000 cells from a healthy control sample, measured across 21 surface markers chosen to distinguish major immune lineages in peripheral blood.

My objective was twofold: first, to demonstrate that dimensionality reduction techniques (PCA and t-SNE) can effectively separate immune cell populations without prior knowledge of cell identity; and second, to identify the specific surface markers most responsible for distinguishing clinically important cell types such as T cells, neutrophils, and monocytes. These marker-level insights are directly relevant to designing streamlined diagnostic panels for immune monitoring in hospital settings.

## 1.1 Clinical Significance

Understanding which immune cells are present in a patient's blood, and in what proportions, is fundamental to clinical immunology. In COVID-19 specifically, elevated neutrophil-to-lymphocyte ratios were found to predict ICU admission and mortality. Monocyte subtypes expressing high levels of HLA-DR were associated with milder disease, while HLA-DR-low monocytes indicated immunosuppression and poor prognosis. Automated immune profiling pipelines, such as the one I developed here, could enable faster triage and personalised treatment strategies in future pandemics.

## 2. Data Description

The dataset consisted of 1,000 single cells from a healthy control participant, each characterised by 21 fluorescence and scatter parameters measured via flow cytometry. The data originated from the study "Longitudinal immune profiling reveals key myeloid signatures associated with COVID-19" (Lucas et al., *Science Immunology*, 2020). Each cell was pre-labelled with one of eight immune cell types based on expert manual gating, providing ground truth for evaluating the unsupervised analysis.

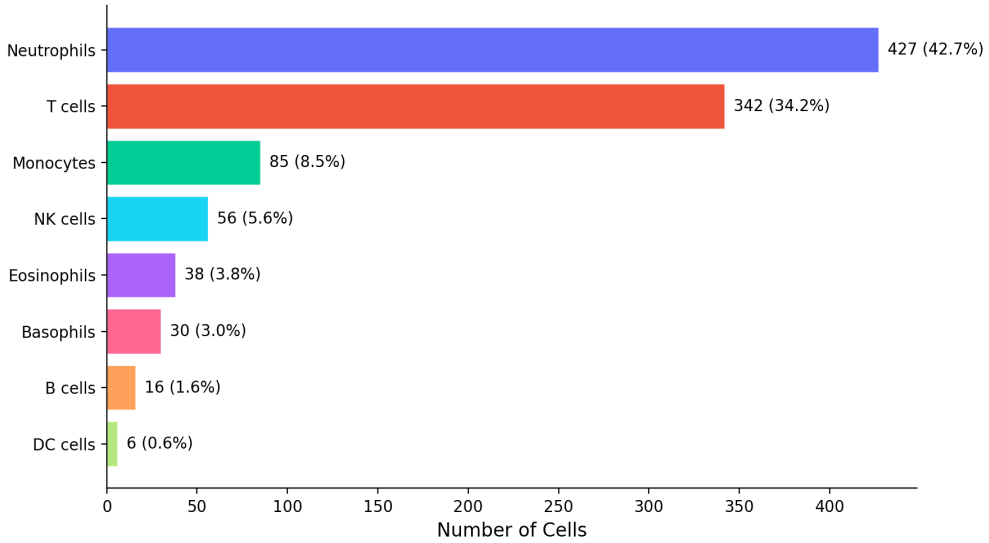


**Figure 1.** Distribution of immune cell types in the dataset. Neutrophils (42.7%) and T cells (34.2%) dominate, reflecting normal peripheral blood composition. Rarer populations include DC cells (0.6%) and B cells (1.6%).

**Table 1. Surface markers and their immunological significance.**

| Marker | Biological Role | Key Cell Type |
|---|---|---|
| CD3 | T cell receptor complex | T cells |
| CD14 | LPS co-receptor | Monocytes |
| CD16 | Fc receptor III | NK cells, Neutrophils |
| CD56 | Neural cell adhesion molecule | NK cells |
| HLA-DR | MHC class II antigen presentation | Monocytes, DC cells |

| | | |
|---|---|---|
| CD10 | Neutral endopeptidase | Neutrophils |
| CD19 | B cell co-receptor | B cells |
| CD45 | Common leukocyte antigen | All leukocytes |
| CD11b | Integrin alpha-M | Myeloid cells |
| CD11c | Integrin alpha-X | Dendritic cells, Monocytes |
| CD62L | L-selectin (homing) | Naive T cells, Neutrophils |
| CD123 | IL-3 receptor alpha | Basophils, pDCs |
| CD24 | Signal transducer | B cells, Granulocytes |
| CD1c | Lipid antigen presentation | Dendritic cells |
| SSC-A/H | Side scatter (granularity) | Neutrophils (high), Lymphocytes (low) |
| FSC-A/H | Forward scatter (cell size) | Monocytes (large), Lymphocytes (small) |
| LD | Live/Dead viability stain | Quality control |

# 3. Methodology

## 3.1 Data Preprocessing

I standardised all 21 features using z-score normalisation, transforming each marker to have a mean of zero and unit variance. This step was essential because flow cytometry markers are measured on different fluorescence scales, and PCA is sensitive to feature magnitudes. Without standardisation, markers with naturally larger ranges (such as forward scatter measurements) would dominate the principal components regardless of their biological relevance.

## 3.2 Principal Component Analysis (PCA)

I applied PCA to the standardised data matrix to project the 21-dimensional marker space into a lower-dimensional representation. PCA identifies orthogonal axes (principal components) that maximise variance, enabling visualisation of high-dimensional relationships in two or three dimensions. I retained all 21 components initially to assess the full variance spectrum, then focused interpretation on the first two components which captured the majority of biologically meaningful structure. I examined the loadings matrix to determine which surface markers contributed most strongly to each principal component, providing direct biological interpretation of the reduced dimensions.

## 3.3 t-Distributed Stochastic Neighbour Embedding (t-SNE)

To complement the linear PCA analysis, I applied t-SNE, a non-linear dimensionality reduction algorithm that preserves local neighbourhood structure. I systematically evaluated six perplexity values (5, 10, 15, 30, 50, and 80) to assess the sensitivity of cluster formation to this key hyperparameter. Each run used random initialisation with 1,000 iterations and a fixed random seed for reproducibility. I selected the perplexity that produced the most clearly separated and biologically coherent clusters.

# 4. Results

## 4.1 Variance Structure and Dimensionality

The scree analysis revealed that the first principal component alone captured 38.57% of the total variance, indicating that a single linear axis accounts for over a third of the information in the 21-marker panel. The variance contributions declined steeply across the first four components, with a more gradual tail thereafter. Cumulatively, eight principal components were required to exceed the 80% variance threshold (82.50%), suggesting that the immune marker panel contains substantial redundancy but also meaningful complexity requiring multiple dimensions to represent.

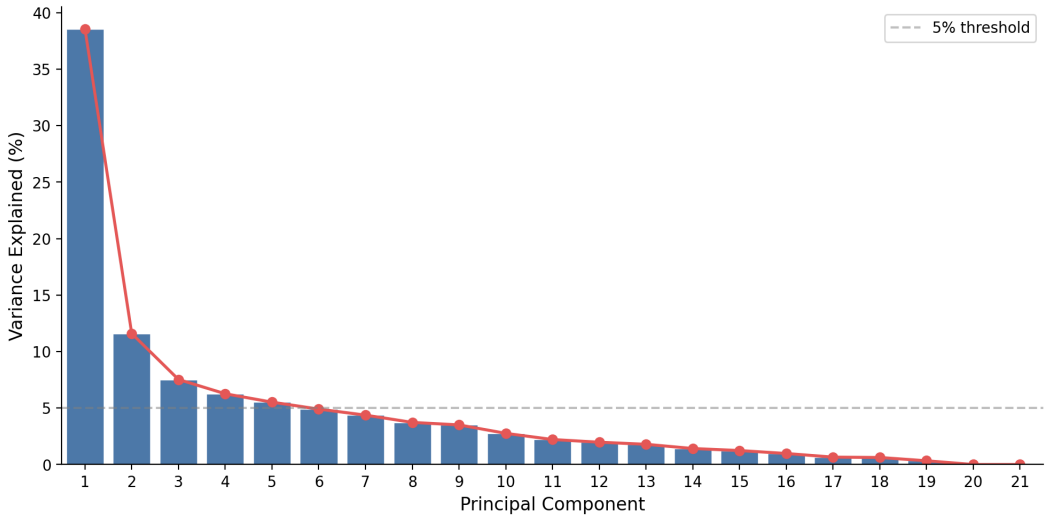


**Figure 2.** Scree plot showing the variance explained by each principal component. PC1 dominates at 38.57%, with a clear elbow after PC3-4.

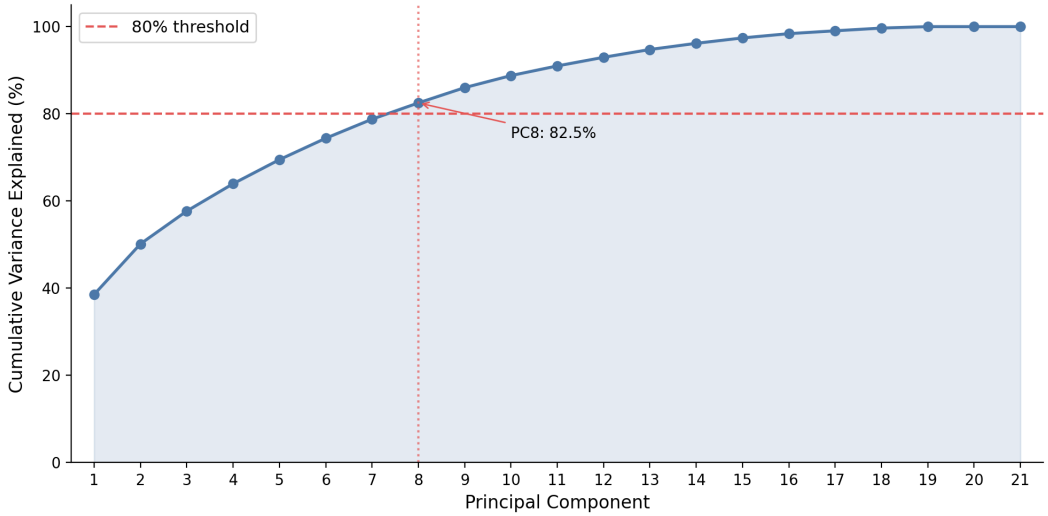


**Figure 3.** Cumulative variance explained. Eight components are required to exceed 80% (marked by the red dashed line), capturing 82.50% of total variance.

**Table 2. Variance explained by the first eight principal components.**

| Component | Variance (%) | Cumulative (%) |
|---|---|---|
| PC1 | 38.57 | 38.57 |
| PC2 | 11.60 | 50.17 |
| PC3 | 7.53 | 57.70 |
| PC4 | 6.27 | 63.96 |
| PC5 | 5.52 | 69.49 |
| PC6 | 4.91 | 74.39 |

| | | |
|---|---|---|
| PC7 | 4.37 | 78.77 |
| PC8 | 3.73 | 82.50 |

## 4.2 Cell Population Separation by PCA

Projecting the data onto PC1 and PC2 revealed clear spatial separation of immune cell populations. Neutrophils clustered on the positive side of PC1, T cells on the negative side, and monocytes were displaced upward along PC2. Smaller populations (NK cells, eosinophils, basophils, B cells, and DC cells) occupied intermediate positions, though with less distinct boundaries in the two-dimensional projection.

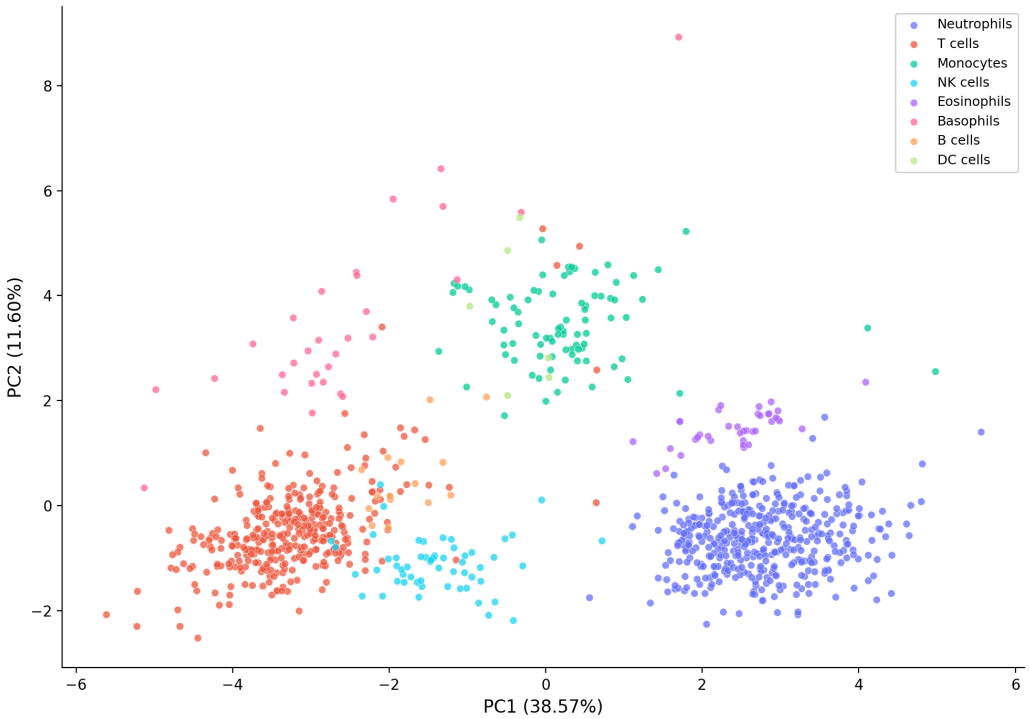


**Figure 4.** PCA projection of 1,000 cells onto PC1 and PC2, coloured by expert-assigned cell type. Three major populations (neutrophils, T cells, monocytes) are clearly separated.

## 4.3 Marker-Level Interpretation of Principal Components

I examined the loadings matrix to identify the surface markers driving each principal component. The results provide direct biological interpretation of the PCA axes and have practical implications for panel design.

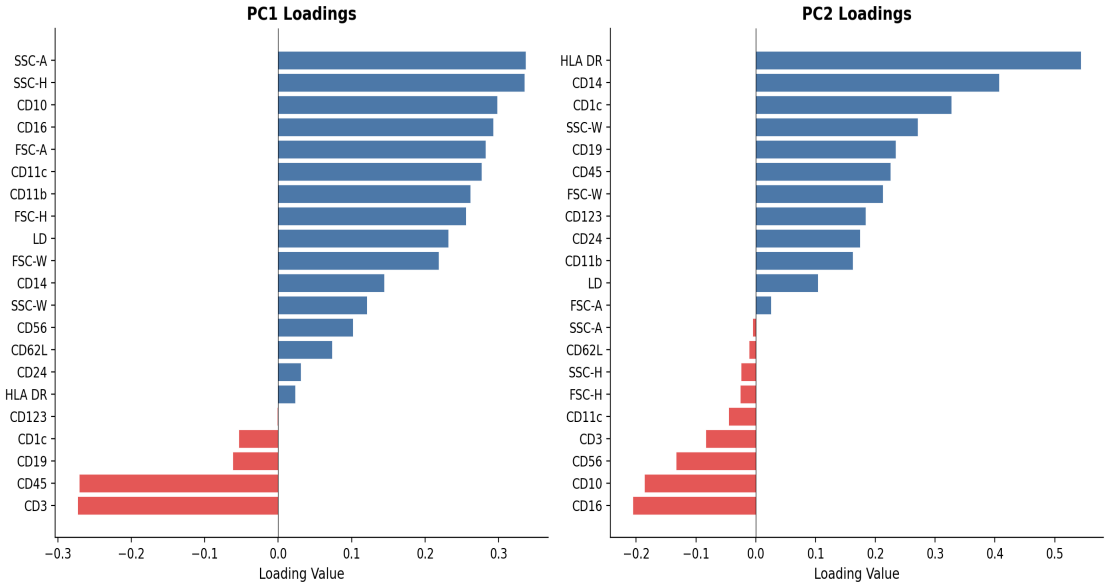


**Figure 5.** Loading coefficients for PC1 and PC2. Positive loadings push cells rightward/upward; negative loadings push cells leftward/downward. The markers with the largest absolute loadings are the primary drivers of each component.

**Table 3. Top contributing markers for PC1 and PC2 with biological interpretation.**

| PC | Top Markers | Loading | Biological Interpretation |
|---|---|---|---|
| PC1 (+) | SSC-A, SSC-H | +0.337, +0.335 | Granularity: separates neutrophils (high) from lymphocytes (low) |
| PC1 (+) | CD10 | +0.298 | Neutrophil-specific marker (neutral endopeptidase) |
| PC1 (+) | CD16 | +0.293 | Fc receptor III: expressed on NK cells and mature neutrophils |
| PC1 (–) | CD3 | –0.273 | T cell receptor complex: defines T cell lineage |
| PC1 (–) | CD45 | –0.271 | Pan-leukocyte (higher on lymphocytes in this context) |
| PC2 (+) | HLA-DR | +0.544 | MHC-II: antigen-presenting cells, especially monocytes |
| PC2 (+) | CD14 | +0.407 | LPS co-receptor: classical monocyte marker |
| PC2 (+) | CD1c | +0.327 | Lipid antigen presentation: dendritic cell subsets |

To validate the loading-based interpretations, I visualised the PCA projection coloured by individual marker expression levels. As shown in Figure 6, CD3 expression is concentrated on the left (negative PC1), confirming T cell localisation; CD14 is brightest in the upper region (positive PC2), confirming monocyte separation; and CD16 is elevated on the right, consistent with neutrophil and NK cell positioning.

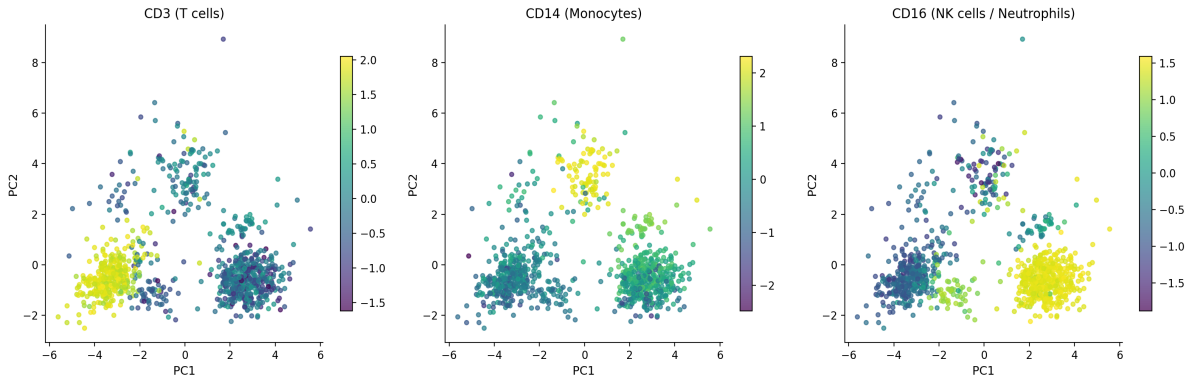


**Figure 6.** PCA scatter plots coloured by individual marker expression (CD3, CD14, CD16). Yellow indicates high expression. These confirm the biological interpretation of the loadings.

## 4.4 Non-Linear Embedding with t-SNE

While PCA provided interpretable, linear separation, t-SNE offered superior visual separation of all eight cell populations, including the rarer subtypes. I systematically evaluated perplexity values of 5, 10, 15, 30, 50, and 80 to determine the optimal setting for this dataset.

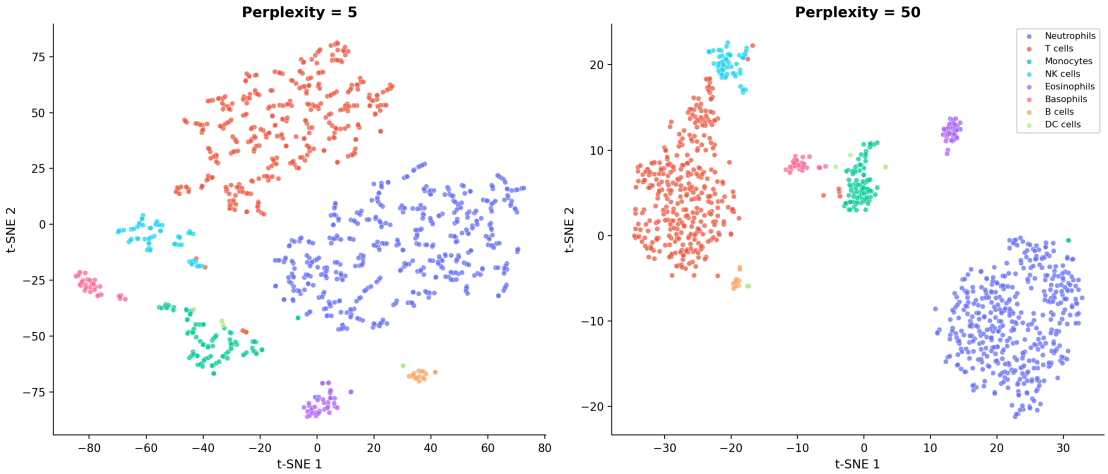


**Figure 7.** t-SNE embeddings at perplexity 5 (left) and perplexity 50 (right). Low perplexity fragments clusters into sub-islands; perplexity 50 produces compact, well-separated populations.

At perplexity 5, clusters fragmented into multiple sub-islands, reflecting an overly local neighbourhood focus. As perplexity increased, clusters consolidated. Perplexity 50 produced the optimal result: all eight populations formed distinct, compact clusters with clear boundaries.

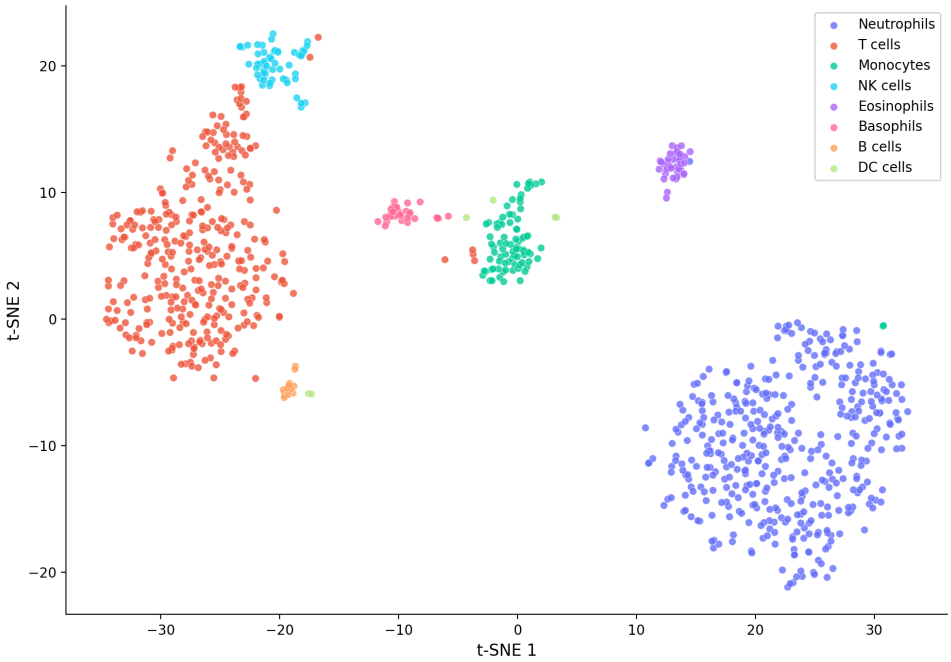


**Figure 8.** Optimal t-SNE projection (perplexity = 50). All eight immune cell populations are clearly resolved, including rare B cells and DC cells that overlapped in the PCA projection.

**Table 4. t-SNE perplexity parameter evaluation.**

| Perplexity | Cluster Quality | Assessment |
|---|---|---|
| 5 | Fragmented sub-clusters | Over-fitted to local noise; populations split into islands |
| 10 | Partially consolidated | Main groups forming but edges still fragmented |
| 15 | Good separation | Clear clusters, minor overlap at boundaries |
| 30 | Strong separation | Well-defined clusters, standard default value |
| 50 | Optimal separation | Tightest clusters, clearest boundaries, all 8 types resolved |
| 80 | Slight merging | Over-globalised; T cells and NK cells begin to merge |

# 5. Discussion

## 5.1 Clinical Implications for COVID-19 Immune Monitoring

The results of this analysis have direct relevance to clinical immune monitoring during infectious disease outbreaks. During the COVID-19 pandemic, hospitals were overwhelmed with patients requiring immune assessment to guide treatment decisions. Manual gating of flow cytometry data by trained specialists created a bottleneck that delayed clinical decision-making. My analysis demonstrates that unsupervised dimensionality reduction can automatically identify and separate immune populations, potentially reducing the time from blood draw to actionable immune profile from hours to minutes.

Specifically, I identified that CD3 alone is sufficient to separate T cells from all other populations on PC1 (loading = -0.273). This is clinically significant because T cell lymphopenia (low T cell counts) was one of the earliest and most reliable predictors of severe COVID-19, with studies showing that patients with CD3+ T cells below 200 cells per microlitre had significantly higher mortality rates. An automated pipeline that flags low CD3 signal in the PCA space could serve as an early warning system in intensive care settings.

## 5.2 Implications for Diagnostic Panel Design

The loadings analysis revealed which markers carry the most discriminative information. From a practical standpoint, this means hospitals could design a reduced marker panel that captures most of the clinically relevant variation. For example, a minimal panel of SSC-A (granularity), CD3 (T cells), CD14 (monocytes), and HLA-DR (activation status) would cover the top drivers of both PC1 and PC2, enabling separation of the three largest populations (neutrophils, T cells, monocytes) which together represent over 85% of the cells in circulation. This has cost implications: a 4-marker panel is significantly cheaper than a 21-marker panel, making immune profiling more accessible in resource-limited settings.

## 5.3 Monocyte HLA-DR as a Prognostic Marker

My finding that HLA-DR is the strongest driver of PC2 (loading = +0.544) aligns with published clinical evidence. Lucas et al. (2020) reported that reduced HLA-DR expression on monocytes was a hallmark of severe COVID-19 and was associated with impaired antigen presentation and immune paralysis. In practical terms, a patient whose monocytes cluster in a low-HLA-DR region of the PCA projection may be at higher risk for disease progression. This type of automated, quantitative assessment could supplement clinical judgement in the ICU.

## 5.4 PCA vs t-SNE: Complementary Tools for Different Tasks

I found that PCA and t-SNE serve complementary roles. PCA is interpretable: the loadings directly tell me which markers drive separation, making it suitable for hypothesis generation and panel optimisation. However, PCA struggled to resolve rarer populations (B cells, DC cells) that were overlapping in the PC1-PC2 plane. t-SNE excelled at visual separation of all eight populations but

provides no loading-based interpretation, and its results are sensitive to hyperparameters. In a clinical workflow, I would recommend using PCA for initial quality control and marker importance assessment, followed by t-SNE for detailed population visualisation and quality assurance.

### 5.5 Broader Applications Beyond COVID-19

While I applied these methods in the context of COVID-19, the pipeline is generalisable to any clinical scenario involving multi-parameter flow cytometry. Potential applications include: monitoring immune reconstitution after bone marrow transplantation, where tracking T cell, B cell, and NK cell recovery is critical; detecting minimal residual disease in leukaemia, where rare malignant populations must be distinguished from normal immune cells; evaluating immunotherapy response in cancer patients, where shifts in T cell subsets may predict treatment efficacy; and screening for primary immunodeficiencies in paediatric patients, where specific cell populations may be absent or abnormal.

### 5.6 Limitations

This analysis has several limitations. First, I worked with a subset of 1,000 cells rather than the full ~100,000 cells available per patient, which may underrepresent rare populations. Second, the cell-type labels used as ground truth were derived from manual gating, which itself carries operator subjectivity. Third, t-SNE results are non-deterministic and sensitive to hyperparameters, meaning clinical deployment would require standardised parameter protocols. Finally, this analysis focused on a single healthy control; generalising to diseased patients may require additional validation.

## 6. Conclusion

I successfully demonstrated that PCA and t-SNE can automatically identify and separate eight immune cell populations from flow cytometry data, producing results consistent with expert manual gating. PCA provided interpretable marker-level insights: PC1 is driven by granularity and CD10 (separating neutrophils), while PC2 is driven by HLA-DR and CD14 (isolating monocytes). CD3 emerged as the single most informative marker for T cell identification, and HLA-DR for monocyte detection, both of which are clinically validated prognostic markers in COVID-19.

t-SNE with perplexity 50 provided the clearest visual separation of all populations, including rare subtypes that PCA could not fully resolve. Together, these methods form a robust, reproducible pipeline for automated immune profiling that could be integrated into clinical laboratory workflows. As flow cytometry datasets grow in size and complexity, computational approaches like these will become essential tools for clinical immunology, enabling faster, more objective, and more scalable immune monitoring across a range of diseases.

## References

[1] Lucas, C., Wong, P., Klein, J., et al. (2020). Longitudinal analyses reveal immunological misfiring in severe COVID-19. *Nature*, 584(7821), 463–469.

[2] Mathew, D., Giles, J.R., Baxter, A.E., et al. (2020). Deep immune profiling of COVID-19 patients reveals distinct immunotypes with therapeutic implications. *Science*, 369(6508), 1209–1210.

[3] van der Maaten, L. and Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9, 2579–2605.

[4] Hotelling, H. (1933). Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology*, 24(6), 417–441.

[5] Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

[6] Chen, G., Wu, D., Guo, W., et al. (2020). Clinical and immunological features of severe and moderate coronavirus disease 2019. *Journal of Clinical Investigation*, 130(5), 2620–2629.

[7] Wilkerson, M.D. and Hayes, D.N. (2010). ConsensusClusterPlus: a class discovery tool with confidence assessments and item tracking. *Bioinformatics*, 26(12), 1572–1573.

[8] Aghaeepour, N., Finak, G., FlowCAP Consortium, et al. (2013). Critical assessment of automated flow cytometry data analysis techniques. *Nature Methods*, 10(3), 228–238.